

ware, and no data can be stored there. All hard disks, including new ones, have bad sectors, which are a normal occurrence during the manufacturing process. The locations of the bad sectors are recorded by the hard disk circuitry, ensuring that they are avoided. However, bad sectors can also form during use, either as a result of improper shutdowns—causing the read / write head to physically scratch the recording surface—or due to normal wear and tear. Newly-developed bad sectors can also be recorded to ensure that they are not accessed the next time. This can be done by performing a “thorough” format of the drive.

2.1.6 Boot Sector

Every partition in a hard disk contains critical data in the first sector, carrying information about loading the operating system. This first sector is called the boot sector. Any partition containing relevant information in the boot sector is termed bootable, and is called the Boot Partition.

2.1.7 BPI

Bits Per Inch refers to the number of bits stored per linear inch of a track.

2.1.8 Cache

Cache is the memory incorporated in the hard disk that stores frequently-accessed data. This saves use of the read/write head, thus extending its life, besides, obviously, improving performance. A 2 MB cache is common in most hard drives, with the highest being 16 MB in the latest, large-capacity drives. The maximum transfer speeds of the interface that is mentioned in the drive specifications refers to the time it takes for the cache to send the data.

2.1.9 CHS

Cylinder Head Sector is a naming scheme used to uniquely address every storage area or sector in a hard disk. Each sector can be uniquely identified, based on the number of the Cylinder, the read / write head operating on that platter, and the number of that sector in that cylinder.